Course milestone M7: First Reproducible Analysis (+ URC Abstract Internal Gate)

HONR 46400 · Evidence-Driven Research

What to submit

One file, `lastname_m07.pdf`. Everything this milestone asks for goes inside it, as sections in the order below.

Submit it on Brightspace, under **Assignments** then **M07**. Brightspace carries the deadline.

Your work lives in a Colab notebook, which has no export-to-PDF button. **Making the PDF you hand in**, at the end of this document, is the one-minute routine that turns the notebook into the file you upload.

This milestone presents **Book Milestone 7: Your first reproducible analysis** (checkpoint, version 1) of the course book, EDR|AI.

What this course adds

The conference submission

Two things leave your hands this week on the conference's calendar rather than on your research schedule, and the book milestone has neither.

1. The abstract the conference publishes. Write the abstract you would submit to the Undergraduate Research Conference, roughly 150 to 250 words: the question, what you measure, the analysis your pipeline executes, the verified result with its uncertainty, and the boundary around the claim. It clears an internal gate at the studio before it goes out. The gate has one rule: the abstract may use only the claims your current verified result licenses. Your result is verified this week and not yet audited, so your wording has to survive next week's robustness work without becoming false.

2. The application that puts your name on the programme. Apply to present a poster through Purdue's Undergraduate Research site. The form wants your title, your abstract with no author name inside the abstract box, five keywords, you as author, in-person poster presentation, your availability on the day, a research category and judging unit, and a faculty mentor of record: Davi Cordeiro Moreira, dcordeir@purdue.edu. An application without a mentor does not stand.

Save the confirmation as a PDF and submit it as your proof of application. Without that confirmation there is no poster session for you to present at, however good the rest of your project is. The deadline belongs to the conference, not to this course, and it is posted on your course platform.

Your Round 01 mentor meeting, confirmed

The meeting you requested at Milestone 4 has to have happened by now, inside the Round 01 window your course platform posts. This milestone collects the proof that it did.

Write one short section into your milestone document naming the day and hour of the meeting, who attended, and the two or three decisions that came out of it. For each decision, say what you changed. Where you kept your course, say that plainly and say why the meeting did not move you. A meeting that changed nothing and cannot say why was a status report.

Record it the day it happens rather than reconstructing it later, and remember that the conversation itself is human work: your AI Research Ledger gets a row only for what you delegated afterwards.

Milestone 7: Your first reproducible analysis

Open the milestone workbook in Colab

This chapter closes Studio 7: Produce a reproducible first analysis. Its lessons each ended at **It is your turn**; what you wrote there arrives here as working material, and this is where the pieces become one artifact you can defend.

What this milestone produces. A route-specific result with an uncertainty statement, a restart-and-run-all verification, an environment record, and a claim-to-output check.

What you bring

You also carry forward the last milestone's artifact: **Milestone 6: Your data and measurement, governed**. This one builds on it, and the chain assumes it is versioned and filed.

Check you are carrying each of these before you start. If one is missing, go back and work that lesson's **It is your turn** first; the practice below assumes it.

- Lesson 22: AI as Programmer: your first computed number, its delegated code read line by line, and the sentence naming what the number does not cover.
- Lesson 23: AI as Analytical Assistant: your headline estimate with its uncertainty statement and cycle log, two independently re-derived numbers, a placebo run where licensed, the clean-restart record, the environment record, and the claim-to-output trace.

The practice

1. Write the analysis you declared, and only that analysis, before looking at anything else.
2. Produce the result with its uncertainty statement, in the form your Contract specified.
3. Restart and run everything from a clean state; if the numbers move, the pipeline is the finding.
4. Record your environment, and check that every claim you plan to make traces to a specific output.

A version, not a pass

Your milestone artifact is a dated, numbered **version** with the reason for the version attached. When later evidence changes it, you write the next version rather than editing the last one, because the sequence of changes is itself part of your research record.

The four rails, here

Four concerns cross every studio in this book. At this milestone they take these specific forms.

- **Ethics, permissions, and data exposure.** What you send to a tool during analysis is a disclosure decision each time.
- **Evidence, provenance, and reproducibility.** Every number in your result traces to a data cell and a line of code.
- **AI activity, verification, and human decisions.** Delegate the writing of code freely; verify every returned number against the data yourself.
- **Uncertainty, claim boundary, and revision history.** A result reported without uncertainty is not yet a result.

Where this milestone sits

You arrived carrying Milestone 6: Your data and measurement, governed. Milestone 8 exists to attack this result; the cleaner your record here, the more the attack can teach you. The chain ends at Milestone 12: Your release and next cycle, where you decide whether your finished research artifact leaves your hands.

What sends you back

Failing to reproduce your own analysis sends you back here before any stress-testing is meaningful.

How this milestone is assessed

Each row scores **0**, **1**, or **2**. **10 points total.**

#	Criterion	0	1	2
1	The milestone artifact exists in full, specific to your own project	absent, or generic text that would fit any project	present but incomplete, or specific in some parts and generic in others	complete and unmistakably about your project

#	Criterion	0	1	2
2	The milestone is written as a dated, numbered version with its reason	no version, or a version with no reason given	versioned, but the reason is decorative rather than an account of what changed	versioned with a reason a reader could use to reconstruct your thinking
3	Every judgment in the artifact is defended by you, not asserted by a tool	conclusions appear with no reasoning, or reproduce a tool's wording	reasoning present but thin where the decision was hardest	each judgment carries a reason you could defend under questioning
4	The four rails are carried in the form this studio requires	one or more rails absent where this studio required them	all rails present but one is nominal	every rail addressed in the form this studio requires
5	The milestone's own product is present and defensible: A route-specific result with an uncertainty statement, a restart-and-run-all verification, an environment record, and a claim-to-output check	the milestone's own product is missing	present but would not survive a careful question	present and defensible as stated

> **! Blocking gate**
>
> No work at or after Studio 4 proceeds past a not-authorized determination. This is not scored and cannot be averaged away.

Your workbook

Open the workbook with the link above. It walks these steps with a cell for each one, ends by writing your milestone version with its reason, and adds the rows this studio contributes to your AI Research Ledger and your Research Dossier.

Making the PDF you hand in

Colab has no export-to-PDF button. You make the PDF by **printing the notebook to a file**, which takes about a minute once you know the preparation steps. Skipping them is the most common way a milestone arrives looking half finished, because a notebook prints what is on the screen and nothing else.

1. Prepare the notebook

1. **Run the whole thing.** `Runtime` ⟶ `Run all`, then wait for it to finish. A cell you never ran prints with nothing underneath it, so an analysis you really did can reach the grader as an empty box.
2. **Expand every collapsed section.** Anything folded shut prints as a heading with nothing beneath it. Open all of them, including any Colab folded for you.
3. **Shorten any output that scrolls.** When Colab puts a scrollbar on an output, or tells you the output is truncated, only the visible part prints. Show the first twenty rows instead of the whole table, or summarise it, then rerun that cell. A reader did not want the whole table anyway.
4. **Read it once, top to bottom.** You are about to freeze it.

2. Print it to a file

`File` ⟶ `Print` opens your browser's print dialog. Set the destination to **Save as PDF**, open the dialog's further settings, and turn on **Background graphics**, so code cells keep their shading and your figures keep their backgrounds. Save it under the file name this milestone asks for.

Wide tables and wide figures are the usual casualty. If something runs off the right edge, switch the layout to **landscape** or reduce the scale, and print again.

3. Check the PDF before you upload it

Open the file you just made and confirm three things: every figure is whole rather than sliced by a page break, no output stops mid-line, and the pages run in order with nothing missing. A PDF is an artifact like any other in this course, so you verify it before your name goes on it.

One route not to take

Ask an AI assistant how to export a Colab notebook and it will very likely hand you `jupyter nbconvert --to pdf`. Do not spend your afternoon on it. Inside Colab that route needs a large LaTeX installation, runs for several minutes, and then either fails or quietly drops the symbols and emoji these notebooks are full of. Printing to a file is the route that works.

That is a small, cheap instance of the rule this whole course runs on. The confident answer is not automatically the correct one, and the way you find out is to check it against what actually happens.